Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2008-II-Q05] | Part A (Q1-Q5)

Answer: 504

1. Extend the legs to meet at E. Since the two base angles sum to ninety degrees, triangle EAD is right at E. Parallelism makes EBC a smaller homothetic right triangle, so E, M, N are collinear.

2. The median to a right triangle's hypotenuse is half that hypotenuse. Thus $EN = 1004$, $EM = 500$, and $MN = 504$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2009-I-Q05] | Part A (Q1-Q5)

Answer: 072

1. In quadrilateral $APCM$, diagonals AC, PM bisect each other, so it is a parallelogram. Thus $AM \parallel CP$, making BLP similar to BAM.

2. Angle-bisector theorem gives $AL/LB = 450/300 = 3/2$. Therefore $LP/AM = BL/BA = 2/5$ and $LP = 72$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2016-II-Q05] | Part A (Q1-Q5)

Answer: 182

1. Put $a = BC_0, b = AC_0, c = AB$. Similarity makes the successive altitudes geometric, with first term ab/c and ratio $a/c < 1$. Hence their sum is $ab/(c-a) = 6(a+b+c)$.

2. Use $(c-a)(c+a) = b^2$ to simplify this equality to $a = 6(b+c-a)$, or $7a = 6(b+c)$.

3. Squaring $6c = 7a - 6b$ and using $c^2 = a^2 + b^2$ gives $13a = 84b$. Coprimality forces $a = 84, b = 13$, and then $c = 85$. These sides satisfy the sum condition, so $p = 182$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2024-II-Q05] | Part A (Q1-Q5)

Answer: 080

1. Let the triangle side lengths be P, Q, R and the hexagon side length s. The three cut-off corner triangles are similar to the large triangle because each cutting side is parallel to the opposite triangle side.

2. Along the side of length P, the two corner pieces have lengths Ps/Q and Ps/R, leaving a central segment of length s. Thus $1 = s(1/P + 1/Q + 1/R)$.

3. Substituting gives $s = 1/(1/200 + 1/240 + 1/300) = 80$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-I-Q07] | Part B (Q6-Q10)

Answer: 923

1. If h is the altitude to BC and r the inradius, similarity gives $DE/20 = (h-r)/h$.

2. Writing area as K, $r = K/s$ and $h = 2K/20$, so $r/h = 20/(2s) = 20/63$. Hence $DE = 20(43/63) = 860/63$, and the sum is 923.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-I-Q09] | Part B (Q6-Q10)

Answer: 035

1. The rectangle cut runs from a vertex to its opposite side and makes a right triangle, which must be similar to the 3–4–5 triangle. Its fitting leg pairs are $(6, 9/2)$ and $(7, 21/4)$; the reversed orientations would exceed the other rectangle side.

2. The residual rectangle trapezoid has two right angles, so the triangle cut must be parallel to a leg rather than the hypotenuse. Matching its acute angle selects a cut parallel to the side of length 3.

3. The trapezoid base ratio, hence the small triangle scale, is $(7 - 9/2)/7 = 5/14$ or $(6 - 21/4)/6 = 1/8$. Their areas are $6(5/14)^2 = 75/98$ and $6(1/8)^2 = 3/32$.

4. The smaller is $3/32$, giving 35.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2015-II-Q07] | Part B (Q6-Q10)

Answer: 161

1. The semiperimeter is 27, so Heron gives triangle area $\sqrt{27 \cdot 15 \cdot 10 \cdot 2} = 90$. Its altitude to BC is $h = 180/25 = 36/5$.

2. Similarity of APQ and ABC gives rectangle height $h(1 - w/25)$. Thus its area is $(36/5)w - (36/125)w^2$, and $\beta = 36/125$.

3. The fraction is reduced, so the requested sum is 161.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2018-II-Q07] | Part B (Q6-Q10)

Answer: 020

1. The first k bands form a similar triangle of area ratio $k/2450$. Thus $P_kQ_k = 5\sqrt{3}\sqrt{k/2450} = \sqrt{6k}/14$.

2. An integer has rational square root exactly when it is a perfect square. The condition $6k$ square is equivalent to $k = 6j^2$ for integer $j \geq 1$, by prime-exponent parity.

3. As $6j^2 \leq 2450$, one has $1 \leq j \leq 20$. The count is 20.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-I-Q06] | Part B (Q6-Q10)

Answer: 090

1. Let F be the foot on KN. Right triangle KLN gives $KF \cdot KN = KL^2 = 784$.

2. Triangles KFO, KMN are similar by their right angles and common angle at K, so $KO \cdot KM = KF \cdot KN = 784$. Hence $KM = 98$ and $MO = 90$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-II-Q07] | Part B (Q6-Q10)

Answer: 715

1. Use barycentric coordinates $\alpha+\beta+\gamma=1$. A parallel chord at level α has length $220(1-\alpha)$, so the three lines are $\alpha=3/4, \beta=3/4, \gamma=7/8$.

2. The new parallel-sided triangle has similarity ratio $|1-(3/4+3/4+7/8)|=11/8$, as seen by subtracting these constant coordinates from a point.

3. Its perimeter is $(11/8)(120+220+180)=715$.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-II-Q13] | Part C (Q11-Q15)

Answer: 069

1. Extend AB beyond B and DC beyond C to meet at P. The first angle relation gives $PA = PD$. The second gives similarity of PCB and PBD.

2. Hence $PC/PB = PB/PD = BC/BD = 3/5$. Since $PB = PD - 8$, solve $(PD - 8)/PD = 3/5$ to get $PD = 20$, $PB = 12$, $PC = 36/5$.

3. Then $CD = PD - PC = 64/5$, so the answer is 69.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2002-II-Q13] | Part C (Q11-Q15)

Answer: 901

1. Assign masses $5, 6, 15$ at A, B, C, matching both side division ratios. Let $F = CP \cap AB$; its mass is 11, giving $CP : PF = 11 : 15$ and $FP/FC = 15/26$.

2. The homothety centered at F with ratio $15/26$ sends C to P and the other two sides to the stated parallels. Hence it sends CAB to PQR.

3. The area ratio is $(15/26)^2 = 225/676$, giving 901.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-II-Q13] | Part C (Q11-Q15)

Answer: 484

1. Cosine law gives $AC = 7$. Let $F = AD \cap CE$; then $ABCF$ is a parallelogram. If B has distance h from AC, F is the same distance on the opposite side.

2. Because $AC \parallel DE$, the triangles with vertex F are similar at ratio $7 : 15$. Convexity puts DE beyond F from AC, so its distance from F is $15h/7$ and from B is $29h/7$.

3. The area ratio is $(7h/2)/[15(29h/7)/2] = 49/435$, giving 484.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2006-II-Q12] | Part C (Q11-Q15)

Answer: 865

1. Triangle ADF has adjacent sides thirteen,eleven and included angle 120°. Its area is $143\sqrt{3}/4$, and cosine rule gives $AF^2 = 433$.

2. Inscribed-angle equalities and the parallel sides give $\triangle CBG \sim \triangle AFD$, with scale ratio $CB/AF = 2\sqrt{3}/\sqrt{433}$.

3. Hence $[CBG] = (12/433)(143\sqrt{3}/4) = 429\sqrt{3}/433$. The requested sum is 865.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2007-I-Q15] | Part C (Q11-Q15)

Answer: 989

1. Write side length s and $AE = t$. Subtract the three corner triangles from ABC: $[DEF] = (\sqrt{3}/4)[5(s-t) + 2t - 10]$. Hence $5(s-t) + 2t = 66$.

2. The sixty-degree angle at E makes AEF similar to CDE by their remaining angles. Thus $t/5 = 2/(s-t)$, so $t(s-t) = 10$.

3. Substitute $s - t = 10/t$ to get $t^2 - 33t + 25 = 0$, giving $t = (33 \pm \sqrt{989})/2$. Then $s = t + 10/t = (231 \pm 3\sqrt{989})/10$. Since $989 = 23 \cdot 43$ is squarefree, the answer is 989.

Source / 来源

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2013-I-Q13] | Part C (Q11-Q15)

Answer: 961

1. Heron gives the initial area ninety. Let $r = 17/25$. The outer corner triangle $B_0C_1C_0$ has similarity ratio r, while $C_0C_1 = 17r$, so the retained AB_1C_1 has linear ratio $t = 1 - r^2 = 336/625$.

2. The removed target triangle at the first step has area $90(1 - r^2 - t^2) = 90r^2t$. Subsequent target triangles have disjoint interiors and successive area ratio t^2.

3. The total is $90r^2t/(1 - t^2) = 90t/(1 + t) = 30240/961$. Since $961 = 31^2$ is coprime to the numerator, $q = 961$.

Source / 来源